

Linux Fundamentals Day 4 — Interview-Oriented Machine Test

Task 1 — User Information Investigation

Execute: `whoami`, `id`, `groups`.

Questions:

1. What is your current username?
`xyaro`
2. Difference between UID and GID?
UID: is the unique id of a user
GID: is the unique id of a group
3. List all groups your user belongs to.
`xyaro adm cdrom sudo dip plugdev users docker`
4. Which group allows administrative access?
`sudo`

Task 2 — Linux User Database

Execute: `cat /etc/passwd`

Questions:

1. Which file stores Linux user information?
`/etc/passwd`
2. Explain `ubuntu:x:1000:1000:Ubuntu User:/home/ubuntu:/bin/bash`. What is the purpose of the login shell field?
These things are the details of a user account (ubuntu).
`ubuntu` – username
`x` – password placeholder
`1000` – UID
`1000` – GID
`Ubuntu` – Full name or other details of the user
`/home/ubuntu` – The path of the user home directory
`/bin/bash` – The login shell of the user

The Login Shell field tells the operating system which command-line interpreter to launch immediately after a user successfully logs in.

Task 3 — Linux Group Database

Execute: `cat /etc/group`

Questions:

1. Which file stores Linux group information?
`/etc/group`
2. What is the purpose of a group in Linux?
It stores the group names and it's GID of the linux system.
3. Explain `docker:x:999:ubuntu`
`docker` is the groupname and `999` is the GID and `ubuntu` is the user who belongs to the `docker` group.

Task 4 — Permission Analysis

Create `employee.txt` and analyze `-rw-r--r--`.

Questions:

Owner permissions, Group permissions, Others permissions, and file type meaning.

Owner Permissions: read and write `rw-`

Group Permission: read only `r--`

Others Permission: read only `r--`

file type: `-` it means it is a regular file

Task 5 — Numeric Permission Conversion

Convert `rwxrwxrwx`, `rwxr-xr-x`, `rw-r--r--`, `rw-----`, `r-xr-xr-x` to numeric values.

Explain why `777` is dangerous and why `600` is recommended.

`777` is dangerous because it gives read, write, execute permissions to user, group and others so anyone can edit, delete, manipulate, execute the file.

`600` is recommended because it gives read and write permission to the user and no permission to groups and others...so it can't manipulate, delete, add, alter the files by no one...only user can edit, view, change the file.

Task 6 — chmod Practical

Apply chmod 777, 644, and 600 to project.txt and explain results.

777 is dangerous because it gives read, write, execute permissions to user, group and others so anyone can edit, delete, manipulate, execute the file.

644 is good safe and secure way of file permission. It gives the group and others read access only and the user can read and write the file....so it helpful for team work and the user can only edit the file ...and group and others can read the file so it will be helpful for their feedbacks and good team coordination.

600 is recommended because it gives read and write permission to the user and no permission to groups and others...so it can't manipulate, delete, add, alter the files by no one...only user can edit, view, change the file.

Task 7 — Script Execution

Create deploy.sh, run chmod +x deploy.sh.

Explain execute permission and DevOps usage.

The execute permission gives the permission to run the file.

In DevOps, giving a script execute permission allows us to run the script directly instead of having to call it through another command. For example, a script such as deploy.sh can be made executable with chmod +x deploy.sh and then run using ./deploy.sh. This is useful because DevOps work involves using Linux scripts to perform tasks repeatedly and consistently.

Task 8 — Directory Permissions

Create DevOpsLab and analyze drwxr-xr-x.

Explain ls -l vs ls -ld and directory execute permission.

ls -l : is using for check the file's details in a long format

ls -ld : is using for check the permissions and details of a directory

Directory execute permission is for enter the directory and traverse the directory

Task 9 — Ownership Management

Use chown and explain difference between chmod and chown.

chmod – change the permission of a directory and file.

chown – change the ownership of a directory and file.

Task 10 — Sudo Verification

Run sudo whoami and sudo -l.

Explain sudo and why it is preferred over root login.

sudo is Super User do. It is use for commands to act the commands like a temporary root.

It doesn't change the terminal to root terminal, it only temporary acts like a root for the particular command. So we don't need to root login, we can use sudo when we need root access without changing the normal login shell to root login shell.

Real DevOps Scenario

deploy.sh → executable

config.env → owner only

application.log → owner write, others read

Assign correct permissions and justify.

deploy.sh → 755: The script needs to be executable, and the owner needs permission to modify it. Other users can run it but cannot modify it.

config.env → 600: The file may contain important configuration information, so only the owner should be able to read or modify it.

application.log → 644: The owner needs to write to the log, while other users only need to read it and should not be able to change it.

Bonus Challenge

Create DevOps-Team directory with:

deploy.sh → 755

app.log → 644

config.env → 600

Verify using ls -l